# Anik Chakraborty

M.Sc. Economics
Indian Institute of Technology, Kanpur

Anik Chakraborty | Anik Chakraborty
anik07chakraborty010203@gmail.com | +91-9002398090

## Academic Qualifications

| Year | Degree/Certificate | Institute | CPI/% |
|---|---|---|---|
| 2025 - Present | M.Sc. Economics | Indian Institute of Technology, Kanpur | 9.77/10 |
| 2025 | B.S.(Hons) Economics | Indian Institute of Technology, Kharagpur | 8.54/10 |
| 2021 | Class XII (WBCHSE) | Ghatal Vidyasagar High School | 98.00% |
| 2019 | Class X (WBBSE) | Ghatal Vidyasagar High School | 97.71% |

## Scholastic Achievements

- Achieved an **All India Rank of 22** among 2,973 candidates who appeared in the **GATE 2025 Economics** examination.
- Secured an **All India Rank of 4,474** among more than 1.41 lakh candidates who took the **JEE Advanced 2021** exam.
- Attained an **All India Rank of 2,907** among 10 lakh candidates who appeared for the **JEE Main 2021** exam nationwide.
- Secured **55th** rank among 65 thousand candidates appeared in the **West Bengal Joint Entrance Examination** in **2021**.
- Ranked **73rd** at the national level in the **ISI B.Stat 2021** entrance examination conducted by the Indian Statistical Institute.
- Secured **State Rank 2** in the **National Talent Search Examination Stage-I 2019** held at the state level in West Bengal.
- Awarded the **NTSE** scholarship after clearing the Stage-II examination in **2019**, conducted by NCERT at the national level.

## Internship Experience

**Research Intern** | TrustBridge Rule of Law Foundation *May'26-Jul'26*

- Conducted an **event-study analysis** on **SEBI-mandated insider trading disclosures** sourced from the **CMIE Prowess database**, cleaning **8,19,457 raw records** to test whether legally disclosed insider trades on the **NSE** carry information about future stock performance.
- Constructed a final event dataset of **26,538 stock-day events** (**11,967** buys, **14,571** sells) across **1,370 NSE-listed stocks**; estimated a **market model** over a **[-260,-61]** day window and computed **Cumulative Abnormal Returns (CAR)** over a **[-60,+60]** event window against the **Nifty** benchmark.
- Found **insider purchases** generated a significant **post-event drift of +3.53%** and **sales -2.33%**, both persistent over **60 trading days**, indicating **genuine private information** rather than a mere disclosure reaction; buying was informative even at small trade sizes, while selling was informative only at scale (**bulk deals >0.5%** of shares outstanding showed **CAR of -5.40%**).
- Recommended **tighter disclosure norms** and enhanced regulatory monitoring for **small-cap** and **bulk** insider trades, where the informational edge was found to be strongest and least eliminated by existing SEBI rules.

## Key Projects

**Using Gossips to Spread Information: A Replication Study** | Course Project, ECO613M, IIT Kanpur *Jan'26-Apr'26*

- Replicated a study using data from **two RCTs** (**213 Karnataka villages**, **521 Haryana villages**) testing whether seeding information through **'gossip nominees'** improves diffusion relative to **random** or **status-based seeding**.
- Estimated **OLS** and **IV regressions** and reproduced the **diffusion-centrality** theoretical framework; found **gossip-seeded villages** received **22% more calls** than randomly seeded villages in Karnataka, and gossip treatment raised **immunization camp attendance by ≈22%** in Haryana, with effects **stable over 13 months**.
- Built an **ML extension** (**Lasso, Ridge, Random Forest, Gradient Boosting**) with **5-fold GroupKFold** cross-validation on **6,466 households** to predict **diffusion centrality** from cheap observable demographics **without network-survey data**, achieving **Spearman** $\rho \approx 0.31$ for ranking accuracy and substantially improving seed selection over random seeding (**+3,472% gain** in total diffusion centrality, leave-one-village-out simulation).

**Role of dividend Policies in Stock Valuation** | BTP, under Dr. Gourishankar S Hiremath, IIT Kharagpur *Jan'25-Apr'25*

- Investigated how different **dividend policies** affect **stock valuation** by analyzing **S&P 500 companies** from **January 2015** to **December 2024**, covering both dividend-paying firms and non-dividend-paying firms across various industry sectors.
- Used **event study methodology** to capture immediate **market reactions** to **dividend announcements** and applied **panel regression** and **machine learning models** (**Random Forest, GBM, SVR**) for long-term performance analysis.
- Found that **dividend increases** led to **positive abnormal returns**, while **cuts** triggered **negative market reactions**; metrics like **dividend yield**, **payout ratio** predicted **risk-adjusted returns** with model **R-square** value exceeding **0.70**.

**Stock Price Prediction:Tata Motors** | BTP, under Dr. Gourishankar S Hiremath, IIT Kharagpur *Jul'24-Nov'24*

- **Forecasted Tata Motors**' daily closing stock price using **Linear Regression**, **Moving Averages**, **Random Forest**, and **Long Short-Term Memory (LSTM)** networks, trained on comprehensive historical stock market data from **2018–2024**.
- **Engineered** multiple time-series features including **30/100-day moving averages**, **volume variability**, and applied **Min-Max scaling** with a **60-day sequence window** to optimize data input for effective LSTM model training and performance.
- **Evaluated** various model performances using **Root Mean Squared Error(RMSE)** and **Mean Absolute Percentage Error(MAPE)**; **LSTM achieved lowest RMSE (30.06)** and **MAPE (2.44%)**, outperforming all other models tested.

## Technical Skills

- **Programming Languages:** Python, C Programming
- **Software and Libraries :** Pandas, NumPy, Matplotlib, Scikit-learn, TensorFlow, MySQL, Stata, LaTeX, MS Excel

## Relevant Courses

| | | |
|---|---|---|
| Probability and Statistics | Statistics for Economics | Indian Economy |
| Derivatives | Econometric Analysis | Economic Data Analysis |
| Financial Management | Corporate Valuation and Restructuring | Equity Research |
| AI & ML | Linear Algebra, Numerical and Complex Analysis | Advanced Calculus |
| Microeconomics | Financial Institutions and Markets | Macroeconomics |
| Development Economics | Programming and Data Structures | Public Finance |
| Economics of Growth | General Equilibrium and Welfare Analysis | International Economics |

## Online Course and Certification

- Python for Data Science and Machine Learning Bootcamp, Udemy. (*Aug, 2025*)